

2018 Spring VLSI DSP Homework Assignment #2

Due date: 2018/4/10

1. Least Mean Square Estimation using Wiener Hopf equation

Given an input signal consisting of 2 frequency components plus a noise term

$$x(n) = \sin(2\pi n/12) + \cos(2\pi n/4) + r(n)$$

$r(n)$ is a random variable with values ranging from -0.05 ~ 0.05 (zero mean with equal probability). Let $y(n)$ be the desired output

$$y(n) = \sin(2\pi(n-2)/12) + 2\cos(2\pi(n-1)/4)$$

Please derive a 16-tap finite impulse response (FIR, with 16 coefficients $b_0 \sim b_{15}$) to predict $y(n)$ based on past inputs $x(n) \sim x(n-15)$. You may use the first 32 input samples, i.e. $x(0) \sim x(31)$ and $y(0) \sim y(31)$ as the training data to calculate the auto-correlation matrix and the cross-correlation vector needed for Wiener Hopf equation.

- Please calculate the auto-correlation matrix and the cross-correlation vector
- Please calculate the linear prediction coefficients $b_0 \sim b_{15}$
- Please calculate the mean (average) square error for $n = 32 \sim 63$
- If the length of training sequence is doubled (from 32 samples to 64 samples), does this yield a better prediction result?

2. LMS adaptive filter design

Please use the LMS criterion to develop a 16-tap adaptive finite impulse response (FIR, with 16 coefficients $b_0 \sim b_{15}$). Use the input signal in problem 1 as the training sequence and determine how many iterations it takes before the filter coefficients converge. The

convergence condition is set as $\frac{\|\mathbf{b}(i) - \mathbf{b}(i-1)\|}{\|\mathbf{b}(i-1)\|} \leq 10^{-4}$, where

$\|\mathbf{b}(i)\| = \sqrt{b_0^2(i) + b_1^2(i) + \dots + b_{15}^2(i)}$ is the norm 2 value of the coefficient vector at the i^{th}

iteration. Set the initial value of the coefficient vector as $\mathbf{b}(0) = \begin{bmatrix} \frac{1}{16} & \frac{1}{16} & \dots & \frac{1}{16} \end{bmatrix}$ and

the step size μ as 2^{-4} . Please compare the result with that of the problem 1 and answer the following question

- Which scheme converges faster
- Which scheme has a smaller prediction error